



In this chapter we study how global parameters of a graph, such as its edge density or chromatic number, can influence its local substructures. How many edges, for instance, do we have to give a graph on n vertices to be sure that, no matter how these edges are arranged, the graph will contain a K^r subgraph for some given r ? Or at least a K^r minor? Will some sufficiently high average degree or chromatic number ensure that one of these substructures occurs?

Questions of this type are among the most natural ones in graph theory, and there is a host of deep and interesting results. Collectively, these are known as *extremal graph theory*.

Extremal graph problems in this sense fall neatly into two categories, as follows. If we are looking for ways to ensure by global assumptions that a graph G contains some given graph H as a *minor* (or topological minor), it will suffice to raise $\|G\|$ above the value of some linear function of $|G|$, i.e., to make $\varepsilon(G)$ large enough. The precise value of ε needed to force a desired minor or topological minor will be our topic in Section 7.2. Graphs whose number of edges is about¹ linear in their number of vertices are called *sparse*, so Section 7.2 is devoted to ‘sparse extremal graph theory’.

sparse

A particularly interesting way to force an H minor is to assume that $\chi(G)$ is large. Recall that if $\chi(G) \geq k + 1$, say, then G has a subgraph G' with $2\varepsilon(G') \geq \delta(G') \geq k$ (Lemma 5.2.3). The question here is whether the effect of large χ is limited to this indirect influence via ε , or whether an assumption of $\chi \geq k + 1$ can force bigger minors than

¹ Formally, the notions of sparse and dense (below) make sense only for classes of graphs whose order tends to infinity, not for individual graphs.

the assumption of $2\varepsilon \geq k$ can. *Hadwiger's conjecture*, which we meet in Section 7.3, asserts that χ has this quality. The conjecture can be viewed as a generalization of the four colour theorem, and is regarded by many as the most challenging open problem in graph theory.

On the other hand, if we ask what global assumptions might imply the existence of some given graph H as a *subgraph*, it will not help to raise invariants such as ε or χ , let alone any of the other invariants discussed in Chapter 1. For as soon as H contains a cycle, there are graphs of arbitrarily large chromatic number not containing H as a subgraph (Theorem 5.2.5). In fact, unless H is bipartite, any function f such that $f(n)$ edges on n vertices force an H subgraph must grow quadratically with n (why?).

dense

edge
density

Graphs with a number of edges about quadratic in their number of vertices are usually called *dense*; the number $\|G\|/\binom{|G|}{2}$, the proportion of its potential edges that G actually has, is the *edge density* of G . The question of exactly which edge density is needed to force a given subgraph is the archetypal extremal graph problem, and it is our first topic in this chapter (Section 7.1). Rather than attempting to survey the wide field of 'dense extremal graph theory', however, we shall concentrate on its two most important results: we first prove Turán's classical extremal graph theorem for $H = K^r$ – a result that has served as a model for countless similar theorems for other graphs H – and then state the fundamental Erdős-Stone theorem, which gives precise asymptotic information for all H at once.

Although the Erdős-Stone theorem can be proved by elementary means, we shall use the opportunity of its proof to portray a powerful modern proof technique that has transformed much of extremal graph theory in recent years: Szemerédi's *regularity lemma*. This lemma is presented and proved in Section 7.4. In Section 7.5 we outline a general method for applying it, and illustrate this by our proof of the Erdős-Stone theorem. Another application will be given in Chapter 9.2. Sections 7.6–7.7 indicate some more advanced tools for applying the regularity lemma.

7.1 Subgraphs

Let H be a graph and $n \geq |H|$. How many edges will suffice to force an H subgraph in any graph on n vertices, no matter how these edges are arranged? Or, to rephrase the problem: which is the greatest possible number of edges that a graph on n vertices can have *without* containing a copy of H as a subgraph? What will such a graph look like? Will it be unique?

extremal

A graph $G \not\supseteq H$ on n vertices with the largest possible number of edges is called *extremal* for n and H ; its number of edges is denoted by

$\text{ex}(n, H)$. Clearly, any graph G that is extremal for some n and H will also be edge-maximal with $H \not\subseteq G$. Conversely, though, edge-maximality does not imply extremality: G may well be edge-maximal with $H \not\subseteq G$ while having fewer than $\text{ex}(n, H)$ edges (Fig. 7.1.1).

 $\text{ex}(n, H)$ 

Fig. 7.1.1. Two graphs that are edge-maximal with $P^3 \not\subseteq G$; is the right one extremal?

As a case in point, we consider our problem for $H = K^r$ (with $r > 1$). A moment's thought suggests some obvious candidates for extremality here: all complete $(r-1)$ -partite graphs are edge-maximal without containing K^r . But which among these have the greatest number of edges? Clearly those whose partition sets are as equal as possible, i.e. differ in size by at most 1: if V_1, V_2 are two partition sets with $|V_1| - |V_2| \geq 2$, we may increase the number of edges in our complete $(r-1)$ -partite graph by moving a vertex from V_1 to V_2 .

The unique complete $(r-1)$ -partite graphs on $n \geq r-1$ vertices whose partition sets differ in size by at most 1 are called *Turán graphs*; we denote them by $T^{r-1}(n)$ and their number of edges by $t_{r-1}(n)$ (Fig. 7.1.2). For $n < r-1$ we shall formally continue to use these definitions, with the proviso that – contrary to our usual terminology – the partition sets may now be empty; then, clearly, $T^{r-1}(n) = K^n$ for all $n \leq r-1$.

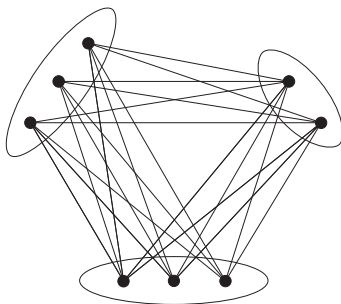
 $T^{r-1}(n)$
 $t_{r-1}(n)$


Fig. 7.1.2. The Turán graph $T^3(8)$

The following theorem tells us that $T^{r-1}(n)$ is indeed extremal for n and K^r , and as such unique; in particular, $\text{ex}(n, K^r) = t_{r-1}(n)$.

Theorem 7.1.1. (Turán 1941)

For all integers r, n with $r > 1$, every graph $G \not\supseteq K^r$ with n vertices and $\text{ex}(n, K^r)$ edges is a $T^{r-1}(n)$.

 $[7.1.2]$
 $[9.2.2]$

We give two proofs: one using induction, the other by a short and direct local argument. While that argument in the second proof is perhaps clever, the first proof is particularly natural and offers more insight.

First proof. We apply induction on n . For $n \leq r - 1$ we have $G = K^n = T^{r-1}(n)$ as claimed. For the induction step, let now $n \geq r$.

Since G is edge-maximal without a K^r subgraph, it has a subgraph $K = K^{r-1}$, with vertices $x_1, \dots, x_{r-1}$ say. From $K^r \not\subseteq G$ alone we have

$$\|G - K\| \leq \text{ex}(n - r + 1, K^r) \quad (1)$$

and

$$\text{Every vertex in } G - K \text{ sends at most } r - 2 \text{ edges to } K. \quad (2)$$

$V_1, \dots, V_{r-1}$ For $i = 1, \dots, r - 1$ let $V_i := \{v \in V(G) \mid vx_i \notin E(G)\}$ (Fig. 7.1.3).

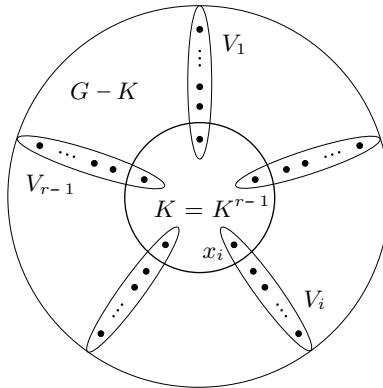


Fig. 7.1.3. Are the sets $V_1, \dots, V_{r-1}$ covering $G - K$ disjoint?
Are they each independent?

Let G' be obtained from K and an $(r - 1)$ -partite Turán graph on $V(G - K)$ by joining its i th partition class completely to $K - \{x_i\}$ for each i . This G' contains no K^r and satisfies both (1) and (2) with equality, in the case of (1) by the induction hypothesis. Hence so does G , because G cannot have fewer edges than G' .

Equality in (1) means, by the induction hypothesis, that $G - K$ is indeed an $(r - 1)$ -partite Turán graph. And its partition classes must be the sets $V_i \setminus \{x_i\}$, since equality in (2) implies with $G \not\supseteq K^r$ that every V_i is independent, and $(r - 1)$ -partite Turán graphs cannot be covered by $r - 1$ sets of independent vertices other than their partition classes. In particular, these classes differ in size by at most 1. Hence G is a Turán graph, too: with classes $V_1, \dots, V_{r-1}$. $\square$

In our second proof of Turán's theorem we shall use an operation called *vertex duplication*. By *duplicating* a vertex $v \in G$ we mean adding to G a new vertex v' and joining it to exactly the neighbours of v (but not to v itself).

vertex
duplication

Second proof. We have already seen that among the complete k -partite graphs on n vertices the Turán graphs $T^k(n)$ have the most edges, and their degrees show that $T^{r-1}(n)$ has more edges than any $T^k(n)$ with $k < r - 1$. So it suffices to show that G is complete multipartite.

If not, then non-adjacency is not an equivalence relation on $V(G)$, and so there are vertices y_1, x, y_2 such that $y_1x, xy_2 \notin E(G)$ but $y_1y_2 \in E(G)$. If $d(y_1) > d(x)$, then deleting x and duplicating y_1 yields another K^r -free graph with more edges than G , contradicting the choice of G . So $d(y_1) \leq d(x)$, and similarly $d(y_2) \leq d(x)$. But then deleting both y_1 and y_2 and duplicating x twice yields a K^r -free graph with more edges than G , again contradicting the choice of G . $\square$

The Turán graphs $T^{r-1}(n)$ are dense: in order of magnitude, they have about n^2 edges. More exactly, for every n and r we have

$$t_{r-1}(n) \leq \frac{1}{2}n^2 \frac{r-2}{r-1},$$

with equality whenever $r - 1$ divides n (Exercise 7). It is therefore remarkable that just ϵn^2 more edges (for any fixed $\epsilon > 0$ and n large) give us not only a K^r subgraph (as does Turán's theorem) but a K_s^r for any given integer s – a graph itself teeming with K^r subgraphs:

Theorem 7.1.2. (Erdős & Stone 1946)

For all integers $r \geq 2$ and $s \geq 1$, and every $\epsilon > 0$, there exists an integer n_0 such that every graph with $n \geq n_0$ vertices and at least

$$t_{r-1}(n) + \epsilon n^2$$

edges contains K_s^r as a subgraph.

A proof of the Erdős-Stone theorem will be given in Section 7.5, as an illustration of how the regularity lemma may be applied. But the theorem can also be proved directly; see the notes for references.

The Erdős-Stone theorem is interesting not only in its own right: it also has a most interesting corollary. In fact, it was this entirely unexpected corollary that established the theorem as a kind of meta-theorem for the extremal theory of dense graphs, and thus made it famous.

Given a graph H and an integer n , consider the number $h_n := \text{ex}(n, H) / \binom{n}{2}$: the maximum edge density that an n -vertex graph can

have without containing a copy of H . Could it be that this critical density is essentially just a function of H , that h_n converges as $n \rightarrow \infty$? Theorem 7.1.2 implies this, and more: the limit of h_n is determined by a very simple function of a natural invariant of H – its chromatic number!

Corollary 7.1.3. *For every graph H with at least one edge,*

$$\lim_{n \rightarrow \infty} \text{ex}(n, H) \binom{n}{2}^{-1} = \frac{\chi(H) - 2}{\chi(H) - 1}.$$

For the proof of Corollary 7.1.3 we need as a lemma that $t_{r-1}(n)$ never deviates much from the value it takes when $r-1$ divides n (see above), and that $t_{r-1}(n)/\binom{n}{2}$ converges accordingly. The proof of the lemma is left as an easy exercise with hint (Exercise 8).

[7.1.2]

Lemma 7.1.4.

$$\lim_{n \rightarrow \infty} t_{r-1}(n) \binom{n}{2}^{-1} = \frac{r-2}{r-1}.$$

□

 r

Proof of Corollary 7.1.3. Let $r := \chi(H)$. Since H cannot be coloured with $r-1$ colours, we have $H \not\subseteq T^{r-1}(n)$ for all $n \in \mathbb{N}$, and hence

$$t_{r-1}(n) \leq \text{ex}(n, H).$$

On the other hand, $H \subseteq K_s^r$ for all sufficiently large s , so

$$\text{ex}(n, H) \leq \text{ex}(n, K_s^r)$$

 s

for all those s . Let us fix such an s . For every $\epsilon > 0$, Theorem 7.1.2 implies that eventually (i.e. for large enough n)

$$\text{ex}(n, K_s^r) < t_{r-1}(n) + \epsilon n^2.$$

Hence for n large,

$$\begin{aligned} t_{r-1}(n)/\binom{n}{2} &\leq \text{ex}(n, H)/\binom{n}{2} \\ &\leq \text{ex}(n, K_s^r)/\binom{n}{2} \\ &< t_{r-1}(n)/\binom{n}{2} + \epsilon n^2/\binom{n}{2} \\ &= t_{r-1}(n)/\binom{n}{2} + 2\epsilon/(1 - \frac{1}{n}) \\ &\leq t_{r-1}(n)/\binom{n}{2} + 4\epsilon \quad (\text{assume } n \geq 2). \end{aligned}$$

Therefore, since $t_{r-1}(n)/\binom{n}{2}$ converges to $\frac{r-2}{r-1}$ (Lemma 7.1.4), so does $\text{ex}(n, H)/\binom{n}{2}$. □

For bipartite graphs H , Corollary 7.1.3 says that substantially fewer than $\binom{n}{2}$ edges suffice to force an H subgraph. It turns out that

$$c_1 n^{2-\frac{2}{r+1}} \leq \text{ex}(n, K_{r,r}) \leq c_2 n^{2-\frac{1}{r}}$$

for suitable constants c_1, c_2 depending on r ; the lower bound is obtained by random graphs,² the upper bound is calculated in Exercise 12. If H is a forest, then $H \subseteq G$ as soon as $\varepsilon(G)$ is large enough, so $\text{ex}(n, H)$ is at most linear in n (Exercise 14). Erdős and Sós conjectured in 1963 that $\text{ex}(n, T) \leq \frac{1}{2}(k-1)n$ for all trees with $k \geq 2$ edges; as a general bound for all n , this is best possible for every T (Exercises 15–17).

*Erdős-Sós
conjecture*

A related but rather different question is whether large values of ε or χ can force a graph G to contain a given tree T as an *induced* subgraph. Of course, we need some additional assumption for this to make sense – for example, to prevent G from just being a large complete graph. The weakest sensible such assumption is that G has bounded clique number, i.e., that $G \not\supseteq K^r$ for some fixed integer r . Then large average degree still does not force an induced copy of T – consider complete bipartite graphs – but large chromatic number might: according to the Gyárfás-Sumner conjecture from Chapter 5.6, there exists for every $r \in \mathbb{N}$ and every tree T an integer $k = k(T, r)$ such that every graph G with $\omega(G) \leq r$ and $\chi(G) > k$ contains T as an induced subgraph.

7.2 Minors

In this section and the next we ask to what extent assumptions about invariants of a graph such as average degree, chromatic number, or girth can force it to contain another given graph as a minor or topological minor.

As a starting question, let us consider the analogue of Turán's theorem: how many edges on n vertices force a K^r minor or topological minor? The qualitative answer is that, unlike for K^r subgraphs where we might need as many as $\frac{1}{2} \frac{r-2}{r-1} n^2$ edges, a number of edges linear in n is enough: it suffices to assume that the graph has large enough average degree (depending on r).

Proposition 7.2.1. *Every graph of average degree at least 2^{r-2} has a K^r minor, for all $r \in \mathbb{N}$.*

Proof. We apply induction on r . For $r \leq 2$ the assertion is trivial. For the induction step let $r \geq 3$, and let G be any graph of average degree at least 2^{r-2} . Then $\varepsilon(G) \geq 2^{r-3}$; let H be a minimal minor of G with $\varepsilon(H) \geq 2^{r-3}$. Pick a vertex $x \in H$. By the minimality of H , x is not iso-

² see Chapter 11